

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: MLA0301

COURSE CODE: Reinforcement learning for Environment

COURSE OUTCOME COVERED

CO2: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)

Assessment Tool Weightage: 30%

Dr G.A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A
Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality.

Solution:

Objective of the Ride-Sharing System

The objective of the ride-sharing system is to efficiently match drivers with passengers by minimizing passenger waiting time, reducing driver travel distance, improving driver utilization, and increasing customer satisfaction. The system should make intelligent matching decisions in real time.

Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning (RL) is suitable because the ride-sharing environment is dynamic. Driver availability, passenger requests, and traffic conditions change continuously. RL learns from experience by receiving rewards for good decisions and penalties for poor decisions, allowing it to improve driver-passenger matching over time.

Agent and Environment

Agent

The **RL agent** is the ride-sharing matching system that selects the most suitable driver for each passenger request.

Environment

The **environment** includes:

- Drivers
- Passengers
- Road network
- Traffic conditions
- GPS locations
- Weather conditions
- Ride requests

State Space (S)

The **state space** represents the current condition of the ride-sharing system.

State Variables:

- Driver location
- Passenger request location
- Driver availability
- Traffic conditions
- Estimated travel time
- Number of ride requests
- Time of day

Action Space (A)

The **action space** represents all possible actions available to the RL agent.

Actions:

- Assign the nearest driver
- Reject a ride request
- Reassign a nearby driver
- Delay assignment for a better match
- Prioritize premium or emergency requests

Reward Function (R)

The reward function encourages efficient ride matching.

Positive Rewards

- Short passenger waiting time
- Short driver travel distance
- Successful ride completion
- High customer satisfaction
- Efficient driver utilization

Negative Rewards

- Long waiting time
- Ride cancellation
- Increased travel distance
- Low customer ratings
- Driver idle time

The RL agent learns to maximize cumulative rewards while minimizing waiting time and improving service quality.

Policy (π)

The **policy** is the strategy used by the RL agent to select the best driver for each passenger request.

Example Policy:

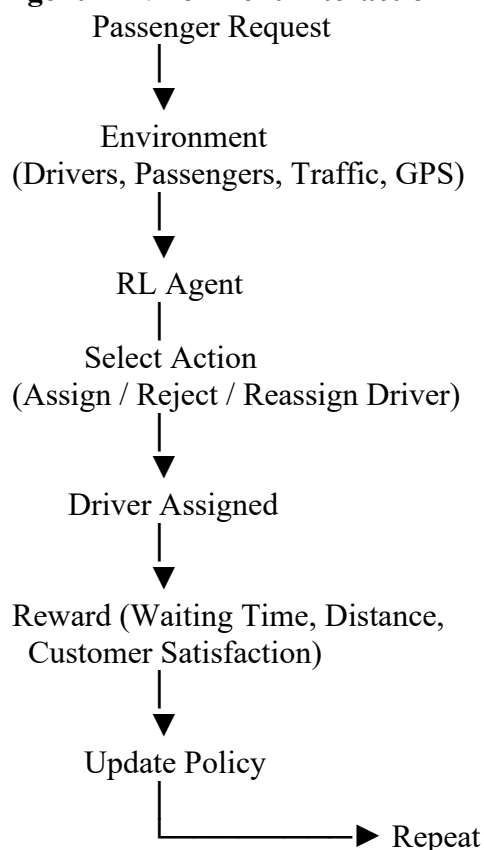
- Assign the nearest available driver if traffic is low.
- Assign the driver with the shortest estimated arrival time during heavy traffic.
- Reassign another nearby driver if the assigned driver cancels.
- Prioritize high-demand areas during peak hours.

The policy improves continuously as the RL agent gains more experience.

Learning Process

1. The RL agent observes the current state.
2. It selects the best driver assignment.
3. The selected action is executed.
4. The system receives a reward based on waiting time, travel distance, and customer satisfaction.
5. The policy is updated using the reward.
6. The process repeats, allowing the RL agent to improve future matching decisions.

Agent–Environment Interaction Diagram



Expected Outcomes

- Reduced passenger waiting time
- Efficient driver utilization
- Shorter travel distance
- Faster ride allocation
- Improved service quality
- Higher customer satisfaction
- Increased ride completion rate
- Better operational efficiency

2. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.

Solution:

Objective of the Robotic Vacuum Cleaner

The objective of the robotic vacuum cleaner is to clean the entire room efficiently while avoiding obstacles such as furniture, walls, and people. The system aims to reduce cleaning time, improve cleaning coverage, and optimize battery usage.

Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning (RL) is suitable because the cleaning environment changes continuously. The robot learns the best cleaning path through trial and error by receiving rewards for successful actions and penalties for mistakes. Over time, it improves its cleaning efficiency without requiring predefined routes.

Agent and Environment

Agent

The **RL agent** is the robotic vacuum cleaner that decides the next movement and cleaning action.

Environment

The **environment** includes:

- Room layout
- Furniture
- Walls
- Obstacles
- Dirty and clean areas
- Charging station

State Space (S)

The **state space** represents the current condition of the robot.

State Variables:

- Current robot location
- Dirty or clean area
- Obstacle locations
- Battery level
- Distance to charging station
- Remaining uncleaned area

Action Space (A)

The **action space** represents the possible actions of the robot.

Actions:

- Move forward
- Move backward
- Turn left
- Turn right
- Start cleaning
- Return to charging station

Reward Function (R)

The reward function encourages efficient cleaning and obstacle avoidance.

Positive Rewards

- Cleaning a new area
- Successfully avoiding obstacles
- Completing room cleaning
- Efficient battery usage

Negative Rewards

- Hitting obstacles
- Revisiting cleaned areas
- Wasting battery power
- Taking unnecessary movements

The RL agent learns to maximize cumulative rewards while cleaning the room efficiently.

Policy (π)

The **policy** is the strategy used by the RL agent to select the best action in each state.

Example Policy:

- Move toward an uncleaned area.
- Turn when an obstacle is detected.
- Continue cleaning until the battery is low.
- Return to the charging station when the battery reaches a critical level.

The policy improves continuously as the robot gains more experience.

Value Function

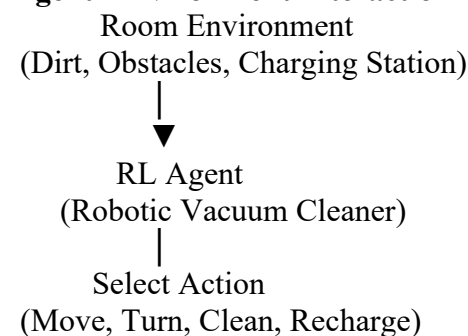
The **value function** estimates the long-term reward of each state and helps the robot choose actions that provide the highest future benefit.

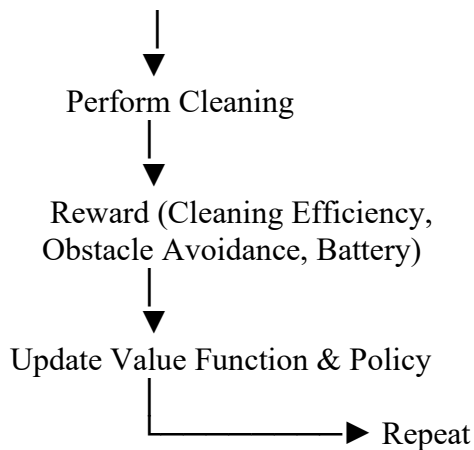
- States leading to efficient cleaning have higher values.
- States causing collisions or repeated cleaning have lower values.
- As the value function is updated, the robot learns shorter cleaning paths, avoids obstacles more effectively, and improves overall cleaning efficiency over time.

Learning Process

1. The robot observes the current state.
2. The RL agent selects the best action based on its policy.
3. The robot performs the action.
4. It receives a reward based on the outcome.
5. The value function and policy are updated.
6. The process repeats until the cleaning task is completed.

Agent–Environment Interaction Diagram





Expected Outcomes

- Efficient room cleaning
- Effective obstacle avoidance
- Reduced cleaning time
- Better battery utilization
- Improved cleaning coverage
- Fewer repeated movements
- Continuous learning and adaptation
- Improved overall cleaning performance

3. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

Solution:

Objective of the Smart Grid System

The objective of the smart grid system is to distribute electricity efficiently by balancing power generation and consumer demand. The system aims to reduce energy wastage, maintain grid stability, improve the use of renewable energy, and minimize operational costs.

Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning (RL) is suitable because electricity demand, renewable energy generation, and grid conditions change continuously. RL learns from real-time interactions and improves electricity distribution decisions over time, making it more effective than fixed rule-based systems.

Agent and Environment

Agent

The **RL agent** is the smart grid controller that decides how electricity should be distributed.

Environment

The **environment** includes:

- Power generation units
- Consumers
- Battery storage systems
- Renewable energy sources (solar and wind)
- Electricity grid

State Space (S)

The **state space** represents the current condition of the smart grid.

State Variables:

- Current electricity demand
- Available power generation
- Battery storage level
- Grid load
- Renewable energy availability
- Time of day
- Electricity price

Action Space (A)

The **action space** represents the possible decisions taken by the RL agent.

Actions:

- Increase electricity supply
- Reduce electricity supply
- Store excess energy in batteries
- Distribute renewable energy
- Purchase electricity from the main grid
- Balance electricity among different regions

Reward Mechanism (R)

The reward mechanism encourages efficient electricity distribution.

Positive Rewards

- Balanced electricity distribution
- Reduced energy wastage
- Efficient use of renewable energy
- Stable power supply
- Lower operational cost

Negative Rewards

- Power outages
- Grid overload
- Energy wastage
- High electricity purchase cost
- Poor load balancing

The RL agent learns to maximize cumulative rewards while ensuring efficient and reliable electricity distribution.

Policy (π)

The **policy** is the strategy used by the RL agent to select the best electricity distribution action.

Example Policy:

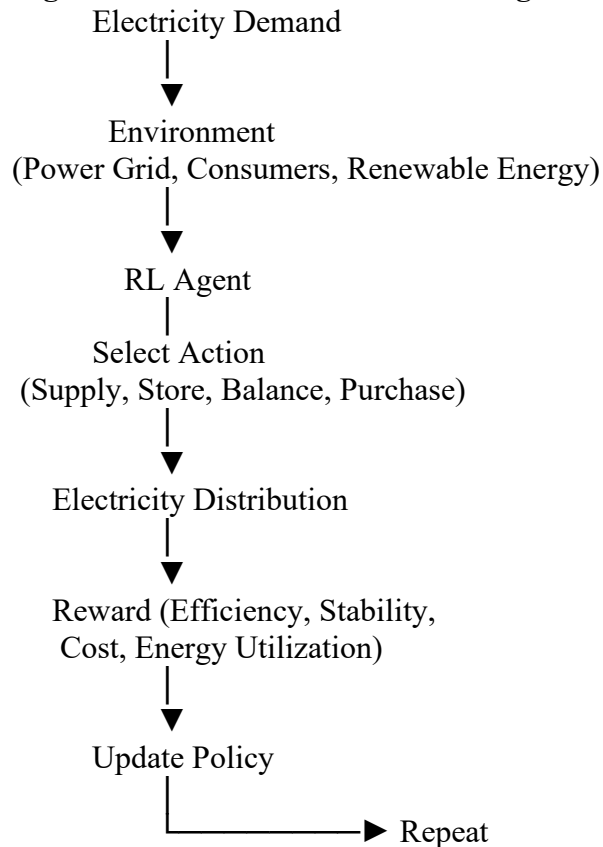
- Supply more electricity during high demand.
- Store excess renewable energy in batteries during low demand.
- Purchase electricity from the main grid only when renewable energy is insufficient.
- Balance the electrical load across different regions.

Learning Process

1. The RL agent observes the current grid state.
2. It selects the best electricity distribution action.
3. The smart grid performs the selected action.

4. A reward is calculated based on grid performance.
5. The policy is updated using the received reward.
6. The process repeats, allowing the system to continuously improve.

Agent–Environment Interaction Diagram



Expected Outcomes

- Efficient electricity distribution
- Reduced energy wastage
- Better utilization of renewable energy
- Lower operational costs
- Improved grid stability
- Reduced power outages
- Continuous adaptation to changing demand
- Improved overall smart grid performance

4. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization.

Solution:

Objective of the Drone Delivery System

The objective of the drone delivery system is to deliver packages safely and quickly by finding the shortest and most efficient route while avoiding obstacles, no-fly zones, and bad weather. The system should minimize delivery time, reduce battery consumption, and improve delivery success.

Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning (RL) is suitable because the drone operates in a dynamic environment where weather, obstacles, and traffic conditions can change frequently. RL enables the drone to learn the best navigation strategy through trial and error, continuously improving its route decisions based on rewards.

Agent and Environment

Agent

The **RL agent** is the drone navigation controller that decides the next movement of the drone.

Environment

The **environment** includes:

- Delivery area
- Obstacles (buildings, trees, birds)
- No-fly zones
- Weather conditions
- GPS navigation system
- Delivery destination

State Space (S)

The **state space** represents the current condition of the drone.

State Variables:

- Current drone location
- Destination location
- Battery level
- Weather conditions
- Obstacle locations
- No-fly zones
- Distance to destination

Action Space (A)

The **action space** represents the possible movements of the drone.

Actions:

- Move forward
- Move left
- Move right
- Move up
- Move down
- Hover
- Land at the destination

Reward Function (R)

The reward function encourages safe and efficient navigation.

Positive Rewards

- Reaching the destination successfully
- Following the shortest route
- Avoiding obstacles
- Low battery consumption
- On-time package delivery

Negative Rewards

- Collision with obstacles

- Entering no-fly zones
- Excessive battery usage
- Delivery delays
- Taking longer routes

The RL agent learns to maximize cumulative rewards by selecting the safest and shortest path.

Policy (π)

The **policy** is the strategy used by the RL agent to choose the best movement at every state.

Example Policy:

- Move toward the destination if the path is clear.
- Change direction when an obstacle is detected.
- Avoid no-fly zones.
- Return to the charging station if the battery level is critically low.

Exploration and Exploitation in Route Optimization

Exploration

- The drone tries new routes to discover shorter or safer paths.
- It helps the system adapt to changing weather and new obstacles.
- Too much exploration may increase delivery time during learning.

Exploitation

- The drone follows the best route learned from previous experiences.
- It reduces travel time and battery consumption.
- Too much exploitation may prevent the discovery of better routes.

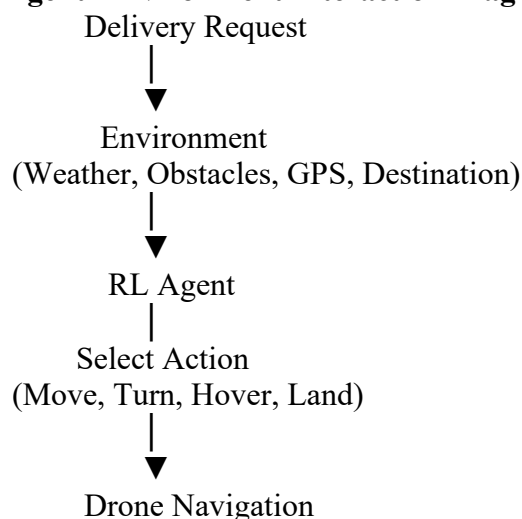
A balance between **exploration** and **exploitation** helps the drone achieve optimal route planning and reliable deliveries.

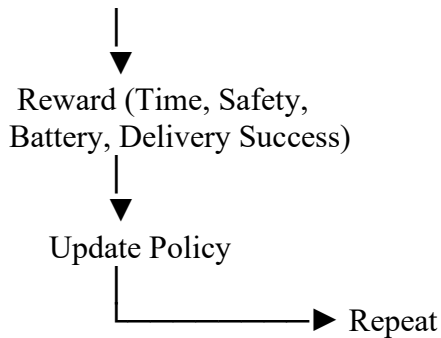
Learning Process

1. The drone observes the current environment.
2. The RL agent identifies the current state.
3. It selects the next movement based on its policy.
4. The drone receives a reward after each action.
5. The policy is updated using the reward.
6. The process repeats until the drone reaches the destination.

Over time, the drone learns the most efficient navigation strategy.

Agent–Environment Interaction Diagram





Expected Outcomes

- Faster package delivery
- Optimized delivery routes
- Reduced battery consumption
- Effective obstacle avoidance
- Safe navigation
- Improved delivery success rate
- Continuous learning and adaptation
- Higher operational efficiency

5. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance.

Solution:

Objective of the Online Advertising System

The objective of the online advertising system is to display the most relevant advertisements to users in order to maximize the **Click-Through Rate (CTR)**, increase conversions, improve user engagement, and generate higher advertising revenue.

Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning (RL) is suitable because user preferences and online behavior change continuously. The RL agent learns from user interactions, such as clicks and purchases, and continuously improves advertisement selection. Unlike rule-based systems, RL adapts to changing user interests and optimizes long-term performance.

Agent and Environment

Agent

The **RL agent** is the advertisement recommendation system that selects the best advertisement for each user.

Environment

The **environment** includes:

- Users
- Advertisements
- User browsing history
- Search queries
- Website or mobile application
- User feedback (clicks or purchases)

State Space (S)

The **state space** represents the current information available before displaying an advertisement.

State Variables:

- User profile
- Browsing history
- Search keywords
- Previous advertisement clicks
- Device type
- Time of day
- User location

Action Space (A)

The **action space** represents all possible advertisement decisions.

Actions:

- Display Advertisement A
- Display Advertisement B
- Display personalized advertisement
- Change advertisement position
- Recommend another advertisement

Reward Function (R)

The reward function encourages advertisements that receive more clicks and improve user satisfaction.

Positive Rewards

- User clicks the advertisement
- Product purchase after clicking
- High user engagement
- Increased conversion rate

Negative Rewards

- Advertisement ignored
- User skips the advertisement
- Low engagement
- High bounce rate

The RL agent learns to maximize total rewards by selecting advertisements with higher click and conversion probabilities.

Policy (π)

The **policy** is the strategy used by the RL agent to choose the most suitable advertisement.

Example Policy:

- Display advertisements based on the user's interests and browsing history.
- Recommend advertisements that previously achieved high click-through rates.
- Explore new advertisements occasionally to discover better-performing options.

Learning Process

The RL agent continuously learns from user interactions.

1. Observe the user's current state.
2. Select an advertisement.
3. Display the advertisement.
4. Receive feedback (click, purchase, or no click).

5. Calculate the reward.
6. Update the policy.
7. Improve future advertisement recommendations.

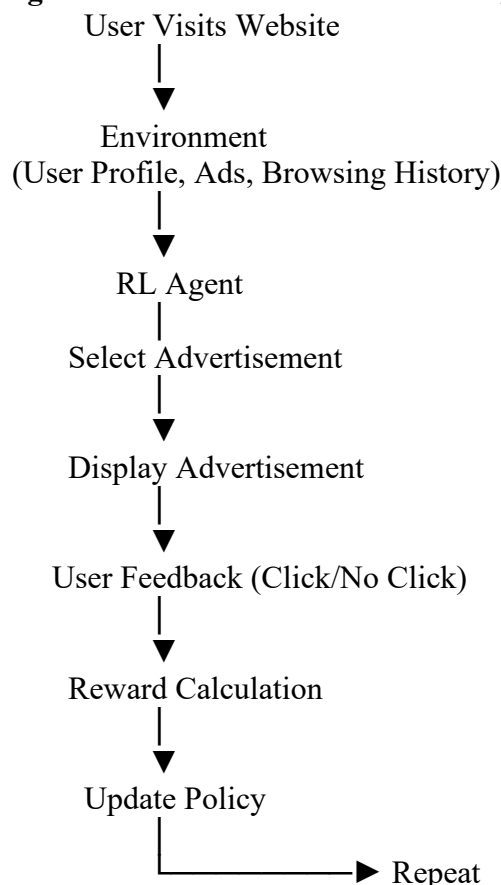
Reward Design Challenges

- **Delayed Rewards:** A user may click an advertisement now but purchase the product later, making it difficult to assign the correct reward.
- **Sparse Rewards:** Many advertisements receive no clicks, providing limited feedback for learning.
- **Balancing CTR and Revenue:** High click rates do not always result in higher sales or profit.
- **Changing User Preferences:** User interests change over time, requiring continuous adaptation.
- **Exploration vs. Exploitation:** The system must balance showing new advertisements (exploration) with showing advertisements known to perform well (exploitation).

Impact of Reward Design on System Performance

- Well-designed rewards improve Click-Through Rate (CTR) and user engagement.
- Accurate rewards help the RL agent learn better advertisement selection strategies.
- Poor reward design can reduce recommendation quality and advertising revenue.
- Continuous reward updates allow the system to adapt to changing user behavior.
- Balanced rewards improve both customer satisfaction and overall business performance.

Agent–Environment Interaction Diagram



Expected Outcomes

- Higher Click-Through Rate (CTR)
 - Increased advertising revenue
 - Better personalized advertisements
 - Improved user engagement
 - Higher conversion rate
 - Continuous learning and adaptation
 - Better customer satisfaction
 - Improved overall system performance
-

Code of Conduct:

*I **Yoga Madha -192425058** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

A handwritten signature in blue ink, appearing to read 'Yogamadha', with a stylized flourish at the end.

Signature of the student

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand